

HAT-P-20b

R-1281

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-20_241124 · created 2026-07-31T14:55:17+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460638.9148 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1401 +/- 0.0073
Transit depth (Rp/Rs)^2	1.96 +/- 0.21 [%]
Orbital Inclination (inc)	88.61 +/- 0.68
Airmass coefficient 1 (a1)	1.0805 +/- 0.0082
Airmass coefficient 2 (a2)	-0.073 +/- 0.023
Scatter in the residuals of the lightcurve fit is	0.76 %
Best Comparison Star	#3 - [480, 367]
Optimal Aperture	3.81
Optimal Annulus	12.8
Transit Duration (day)	0.0909 +/- 0.003

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1401 $\pm$ 0.0073 vs 0.1284 (deviation 1.6 σ)
Duration fit vs published	0.0909 d vs 0.0775 d (deviation 4.47 σ)
Mid-time O-C	-2.8 min
Depth SNR	19.2
Residual scatter	0.76 %

Light curve

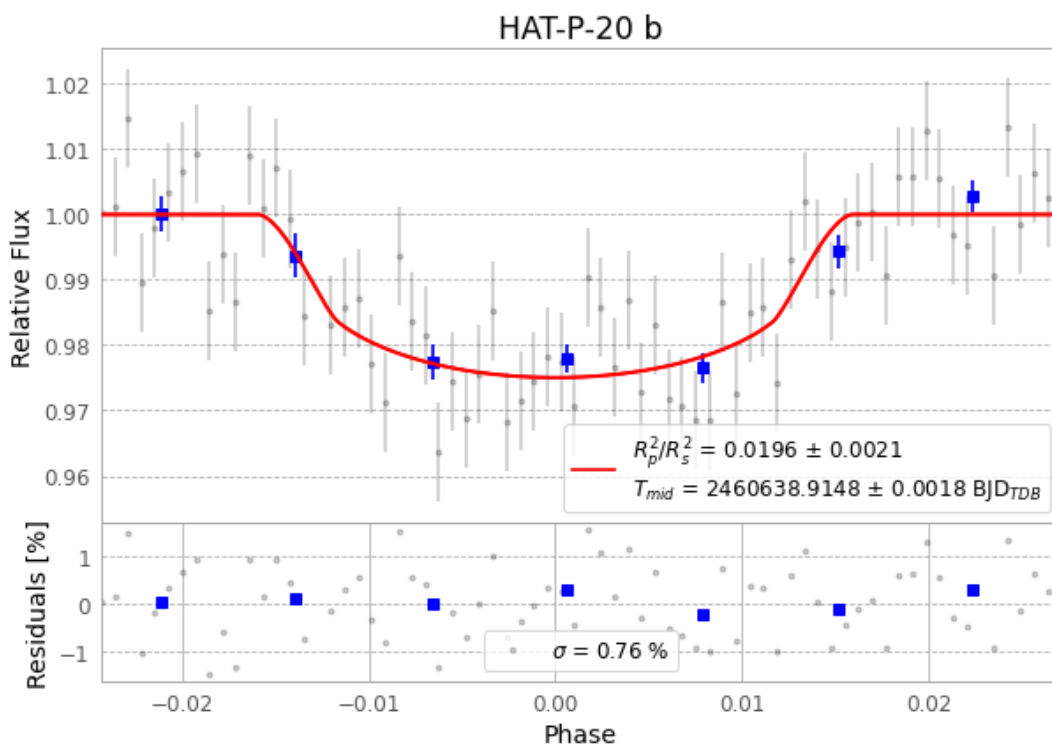


Figure 1 — FinalLightCurve_HAT-P-20 b_2024-11-24.png (SHA-256 788747d95d8191d5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [385, 240], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 86c2899414996cfd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

72 FITS frames (47 MB), HATP-20241124080915.FITS → HATP-20241124114215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-20-241124.log (7f05a1728f7d...), FinalLightCurve_HAT-P-20 b_2024-11-24.png (788747d95d81...), FinalLightCurve_HAT-P-20 b_2024-11-24.csv (dd38c6beaa4e...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 14:55Z.